



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -1

CO1_AT1

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1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

ANSWER :

1. 150 MB file transfer

- File size = 150 MB = $150 \times 1024 \times 1024 = 157,286,400$ bytes
- Segment size = 1500 bytes

Number of segments:

- $157,286,400 \div 1500 \approx 104,857.6$
- Total segments = 104,858

Frames transmitted:

- Each segment becomes one frame.
- Total frames = 104,858

Data Link overhead:

- Overhead/frame = 40 bytes
- Total overhead = $104,858 \times 40 = 4,194,320$ bytes (~4 MB)

Transmission time:

- Total transmitted data = $157,286,400 + 4,194,320 = 161,480,720$ bytes
- In bits = 1,291,845,760 bits
- Bandwidth = 50 Mbps

Transmission time = $1,291,845,760 / 50,000,000 = 25.84$ seconds

Reliability:

- Transport Layer: Segmentation, acknowledgments, retransmissions.
- Data Link Layer: Framing, error detection (CRC), flow control.

2.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

ANSWER :

Sliding Window (48 frames)

- Window size = 6
- Total frames = 48

Windows required:

- $48 \div 6 = 8$ windows

Retransmissions:

- 1 frame lost/window
- Total retransmissions = $8 \times 1 = 8$ frames

Flow Control:

- Prevents sender from overwhelming the receiver.
- Improves throughput and reduces congestion.

3.A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

ANSWER :

IP Fragmentation

- Packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Available payload per fragment:

- $1500 - 20 = 1480$ bytes

Fragments:

- $12,000 \div 1480 \approx 8.11$
- Total fragments = 9

Header overhead:

- $9 \times 20 = 180$ bytes

Network Layer Role:

- Performs fragmentation.
- Uses identification, offset, and MF (More Fragments) fields.
- Destination reassembles fragments correctly.

4.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

ANSWER :

Sliding Window (Same as Q2)

- Windows = $48 \div 6 = 8$
- Retransmissions = $8 \times 1 = 8$

Flow Control:

- Maintains efficient communication.
- Avoids packet loss due to receiver buffer overflow.

5.A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

ANSWER :

Session Layer Synchronization

- File size = 600 MB
- Checkpoint every 60 MB
- Failure at 390 MB

Checkpoints before failure:

- $390 \div 60 = 6.5$
- Completed checkpoints = 6 (at 60,120,180,240,300,360 MB)

Data retransmitted:

- Failure after 390 MB
- Last checkpoint = 360 MB
- Data to retransmit = $390 - 360 = 30$ MB

Importance:

- Synchronization checkpoints allow recovery from failures.
- Prevents retransmission of the entire file.

6.Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

ANSWER :

Presentation Layer Compression & Encryption

- Original size = 900 MB

Compressed size:

- Reduction = 40%
- $900 \times 0.60 = 540$ MB

Encrypted size:

- Overhead = 5%
- $540 \times 1.05 = 567 \text{ MB}$

Total reduction:

- Reduction = $900 - 567 = 333 \text{ MB}$
- Percentage = $(333/900) \times 100 = 37\%$

Presentation Layer Functions:

- Compression improves efficiency.
- Encryption provides confidentiality and security.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

ANSWER :

FTP Transfer

- Data = 3 GB = 3072 MB
- Request size = 3 MB

Requests generated:

- $3072 \div 3 = 1024 \text{ requests}$

Transmission time:

- 3 GB = 24,576 Mb
- Throughput = 120 Mbps

Time = $24,576 \div 120 = 204.8 \text{ seconds } (\sim 205 \text{ s})$

Application Layer Services:

- Provides FTP services.
- Handles user authentication, file transfer, and error reporting.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

ANSWER :

Layer Delay	Layer Delay
Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Total delay:

- $4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$

Layer Contributions:

- Application: User services.
- Transport: Reliability.
- Network: Routing.
- Data Link: Framing.
- Physical: Bit transmission.

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

ANSWER :

Network Efficiency

- Useful data = 80 MB = 83,886,080 bytes
- Frame size = 1600 bytes
- Overhead/frame = 80 bytes

Useful data/frame:

- $1600 - 80 = 1520$ bytes

Frames required:

- $83,886,080 \div 1520 \approx 55,188.21$
- Total frames = 55,189

Total overhead:

- $55,189 \times 80 = 4,415,120$ bytes

Transmission efficiency:

- $(1520/1600) \times 100 = 95\%$

Effect of Overhead:

- Excessive overhead lowers efficiency and increases transmission time.

10.A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

ANSWER :

Hospital Medical Record Transfer

- Original file = 240 MB

Step 1: Compression

- Compression = 30%
- Compressed size = $240 \times 0.70 = 168 \text{ MB}$
- 168 MB = 176,160,768 bytes

Step 2: Segmentation

- Segment size = 1400 bytes

Segments:

- $176,160,768 \div 1400 \approx 125,829.12$
- Total segments = 125,830

Step 3: Data Link Overhead

- Overhead/frame = 60 bytes

Total overhead:

- $125,830 \times 60 = 7,549,800 \text{ bytes}$

Step 4: Final Data Transmitted

- Total = $176,160,768 + 7,549,800$
- = 183,710,568 bytes
- $\approx 175.2 \text{ MB}$

OSI Layer Cooperation:

- Presentation Layer: Compression and security.
- Transport Layer: Segmentation and reliability.
- Data Link Layer: Framing and error detection.
- Physical Layer: Actual transmission of bits across the network.